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Selection Committee
School of Electrical Engineering and Computer Science
Faculty of Engineering, Architecture and Information Technology
The University of Queensland
St Lucia QLD 4072, Australia

Dear Selection Committee,

Application for Lecturer in Electrical Engineering / Biomedical Engineering, R-63679

I am writing to apply for the Lecturer position in Electrical Engineering / Biomedical Engineering (R-63679) advertised by the School of Electrical Engineering and Computer Science. I am currently a Research Assistant at the Power Electronics and Machines Centre (PEMC), University of Nottingham, having completed my PhD in Electrical Engineering at Southeast University, Nanjing, in September 2025. The combination of broad teaching across the electrical engineering curriculum and a research home in a School with strong Imaging, Sensing and Biomedical Engineering activity is an excellent fit for the methodology-led research and teaching programme I want to build at UQ.

My research develops multiphysics modelling and AI-driven surrogate techniques for the analysis, design and optimisation of magnetic and electromagnetic devices. The work has three methodological strands: (i) strongly coupled electromagnetic, thermal, fluid and mechanical finite-element analysis; (ii) high-resolution lumped-parameter thermal networks combined with oil-cooling fluid-solid solvers; and (iii) operator-learning neural surrogates for the magnetic field that achieve design-loop accuracy at orders-of-magnitude lower cost than conventional numerical analysis. To date I have authored 22 journal papers (including 17 in IEEE Transactions), 17 peer-reviewed conference papers, and hold 9 patents. The work has been recognised by the 2025 1st Prize in the IEEE TEC Electric Vehicle Design Competition, the 2025 2nd Prize Paper Award of *IEEE Transactions on Transportation Electrification*, the 1st Prize in the IEEE IAS CMD Thesis Contest, and best-paper recognitions at IEEE-affiliated conferences. My principal application domain to date has been high power-density electric drive systems for electrified transport. The underlying methodology, including multiphysics coupling, magnetic-field forward modelling and learned operators, is application-agnostic and is well placed to support cross-disciplinary collaboration with the School's imaging and sensing groups, for example on electromagnetic device design for MRI gradient and shimming systems, electromagnetic field analysis for non-invasive stimulation devices, and computationally efficient surrogates for inverse problems in electromagnetic imaging.

A distinguishing feature of my profile is full industrial development experience. In partnership with Inovance, a leading EV traction-machine supplier, I have analysed, designed, manufactured and tested two advanced oil-cooled PM traction machines: an 18 000 rpm 113 kW machine for an A0-class vehicle (138 Nm peak torque, 7.78 kW/kg power density, 95.3% maximum efficiency) and a 20 000 rpm 270 kW machine for an A-class vehicle (380 Nm peak torque, 7.87 kW/kg power density, 97.4% maximum efficiency). I led the multi-physics design, produced engineering drawings for both electromagnetic and structural components, and characterised both prototypes on an AVL dynamometer system, extracting voltage and current waveforms, key performance indicators and full efficiency maps. I have also developed an in-house finite-element code package for coupled electromagnetic, structural-modal and mechanical-stress analysis of electrical machines, which is now being applied to magnet-free synchronous machine development at

PEMC Nottingham. This combination of academic and industrial credibility positions me well to pursue ARC Discovery, DECRA and Linkage funding, and to attract direct industrial partnerships once based at UQ.

I am prepared to teach into several of the curriculum areas listed in the position description, in particular **introductory programming for engineering students**, **intermediate systems programming**, **embedded and digital systems** across the undergraduate progression, and **electrical engineering team and individual capstone projects**. My day-to-day technical toolchain spans Python, MATLAB/Simulink, C and C++ for embedded platforms (TI C2000, STM32, FPGA), and machine-learning frameworks for operator-learning surrogates. I have designed and implemented a three-phase full-bridge converter and its control algorithm for testing flux-modulation machines, and have built an in-house Rust-based MCU plotting and data-acquisition tool used in our laboratory work. These experiences map directly onto the laboratory and project components of the courses listed above and onto the supervision of capstone projects in EV electrification, motor-drive prototyping and embedded control.

My direct teaching experience is consistent with my career stage. At Southeast University I served as teaching assistant for the postgraduate course *Integrated Practice of Electrical Machine Systems*, where I prepared and delivered tutorial slides for selected sessions, ran daily Q&A and consultation hours, reviewed student final-project outline proposals, and provided structured mark recommendations to the course coordinator. I have presented at international conferences in Sweden, the United States, Italy, Australia and China, which has built the audience-management and live-questioning skills that translate directly to lecture-room and capstone-review settings. I am also Co-Organiser and Co-Chair of a Special Session at the 2026 International Conference on Electrical Machines (ICEM 2026, Covilhã, Portugal) on *Next-Generation Electrified Transport: Advances in Electric Drive Systems*, jointly with Dr Lingyun Shao (Nanjing University of Aeronautics and Astronautics), which involves recruiting an international author cohort, coordinating peer review with the conference chairs, and chairing the live session. I would welcome UQ's standard induction support for new Level B lecturers as I take on full course-coordination responsibility.

I have attached my CV and detailed responses to the "About You" selection criteria. Thank you for your consideration. I would be very pleased to discuss how I could contribute to the School's teaching programme and research strategy.

Yours sincerely,

Wentao Zhang